

Salvatore Barbagallo

Bioinformatician | Statistical Genetics · Transcriptomics · NGS

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PROFILE

Bioinformatician with an MSc in Bioinformatics and seven years in regulated clinical laboratories, two of them in high-throughput clinical NGS. I build reproducible analyses and pipelines in R, Python and Nextflow, most recently proteome-wide Mendelian randomisation and colocalisation for Parkinson's disease target prioritisation. I know how sequencing data is generated, where it fails QC, and what audit-ready documentation requires.

BIOINFORMATICS PROJECTS

Proteome-wide cis-MR and colocalisation scan for Parkinson's disease

2026

Nextflow · Python · Docker · pytest · UKB-PPP · [GitHub repository](#)

- Scaled my dissertation from one protein to a Nextflow pipeline built for the ~2,900-protein UKB-PPP panel: per-protein cis-MR (Wald, IVW, MR-Egger, weighted median), colocalisation, FDR control and evidence tiers.
- Reproduces the dissertation's R results (coloc.abf, IVW, Wald ratio) to eight significant figures; 14 automated tests and a test profile run in CI.

GPNMB and Parkinson's disease: Mendelian randomisation and colocalisation

2026

MSc dissertation · R · TwoSampleMR · coloc · susieR · OpenGWAS · [GitHub repository](#)

- SomaScan cis-instrument rs5850: OR 1.32 (95% CI 1.19–1.46; $P = 2.6 \times 10^{-7}$); coloc.abf supported a shared signal (PP.H4 = 0.94), checked for prior sensitivity and with SuSiE.
- Heterogeneous Olink instruments were non-significant under random effects, so I concluded prioritisation rather than causality.

GPNMB in Parkinson's substantia nigra: single-nucleus RNA-seq re-analysis

2026

Python · Scanpy · PyDESeq2 · R/speckle · [GitHub repository](#)

- Re-analysed 83,484 nuclei from 29 donors (GSE243639) and localised GPNMB mainly to microglia.
- Donor-level pseudobulk DE: GPNMB ~2-fold higher in PD microglia (log2FC 1.03, $P = 0.009$), not significant after transcriptome-wide FDR (0.22).

Containerised RNA-seq pipeline, validated on real data

2026

Nextflow DSL2 · Docker · STAR · Salmon · DESeq2 · [GitHub repository](#)

- FASTQ to differential expression in one command, with STAR + featureCounts or low-memory Salmon routes and configurable DESeq2 designs.
- Airway dataset (GSE52778, paired donors): 94–95% mapping, 615 DE genes, and 8/8 known glucocorticoid-response genes recovered ($\text{padj} < 1e-9$).

RegulonML: predicting regulatory variant effects in unseen promoters and enhancers

2026

Python · scikit-learn · JASPAR · MPRA · [GitHub repository](#)

- Modelled 41,724 variants in 21 regulatory elements from sequence alone, scoring binding-site gain and loss for 1,019 JASPAR motifs under leave-one-locus-out CV.
- Motif features doubled held-out correlation (Spearman 0.07 → 0.13); showed that a random split inflates it to 0.31.

M. tuberculosis WGS drug-resistance workflow

Galaxy · BWA-MEM2 · bcftools · SnpEff · IGV · [GitHub repository](#)

- SRA reads to annotated variants for 6 clinical isolates ($\geq 52\times$ depth at every target locus); called rpoB and katG resistance mutations, confirmed in IGV, identifying 4 MDR-TB isolates.

EXPERIENCE

Bioinformatician, Scientific AI Evaluation · Mercor (contract, remote) Jan 2026 – present

- Authored 106 accepted bioinformatics benchmark tasks for a frontier model: hidden Python functions implementing cited genomics rules, with unit tests.
- Calibrated difficulty from model trajectories so textbook answers scored below 50%; 80% first-pass approval in review.
- Currently evaluating AI agent runs on preclinical drug R&D tasks and comparing coding models on large codebases.

Specialist Biomedical Scientist, Stem Cell Laboratory · UCL Hospitals NHS Trust Sep 2021 – Dec 2025

- Built Python/Excel tracking tools that cut inventory errors by ~30% and saved ~10 hours a week.
- Kept audit-ready traceability for 11 clinical and ATMP trials, including CAR-T, and digitised SOPs and QA documentation.
- Processed and cryopreserved PBSC, bone marrow and CD34+ products; produced flow cytometry data for clinical decisions.

Laboratory Scientist, Clinical Genomics · CooperGenomics, London Jul 2019 – Sep 2021

- Prepared and QC'd NGS libraries for PGT-A, PGT-SR and PGT-M, at 500+ samples a week to clinical turnaround targets.
- Programmed and validated Mosquito HV and Dragonfly liquid handlers; wrote and reviewed SOPs to ISO standards.

Biomedical Laboratory Assistant, Cytology · Leicester Royal Infirmary Dec 2018 – Jun 2019

EDUCATION

MSc Bioinformatics · Atlantic Technological University (remote) Sep 2025 – Sep 2026

Dissertation: Prioritising GPNMB as a Parkinson's disease-relevant protein using MR and colocalisation

MSc Cell and Gene Therapy · University College London Sep 2021 – Sep 2023

Dissertation: Haematopoietic potential of human amniotic fluid stem cells

BSc Biomedical Science · University of Catania, Italy Sep 2014 – Oct 2017

TECHNICAL SKILLS

Programming: Python (pandas, Scanpy, PyDESeq2, scikit-learn, pytest), R (TwoSampleMR, coloc, susieR, DESeq2, tidyverse), Bash, SQL

Workflows: Nextflow (DSL2), Docker, Galaxy, Conda, Git, GitHub Actions, Linux

NGS: FastQC, Cutadapt, STAR, Salmon, featureCounts, BWA-MEM2, samtools, bcftools, SnpEff, MultiQC, IGV

Statistical genetics: two-sample and cis-MR, Bayesian colocalisation, SuSiE, LD clumping, FDR control, GWAS / pQTL / eQTL

Single-cell and ML: donor-level pseudobulk DE, composition testing, gradient boosting, grouped cross-validation

Languages: Italian (native), English (fluent), Portuguese (fluent), Spanish (advanced)